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Reputation Dynamics in Online Marketplaces: Evidence from Amazon Product Reviews

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CONTENT

1.1 Research Questions

Question 1

How do ratings and review volume affect product popularity

Question 2

Does the effect of ratings depend on review stage?

Question 3

Are consumers more sensitive to reputation signals when information is scarce?

The research purpose of this project is to investigate how algorithmic reputation signals affect product outcomes in online marketplaces.

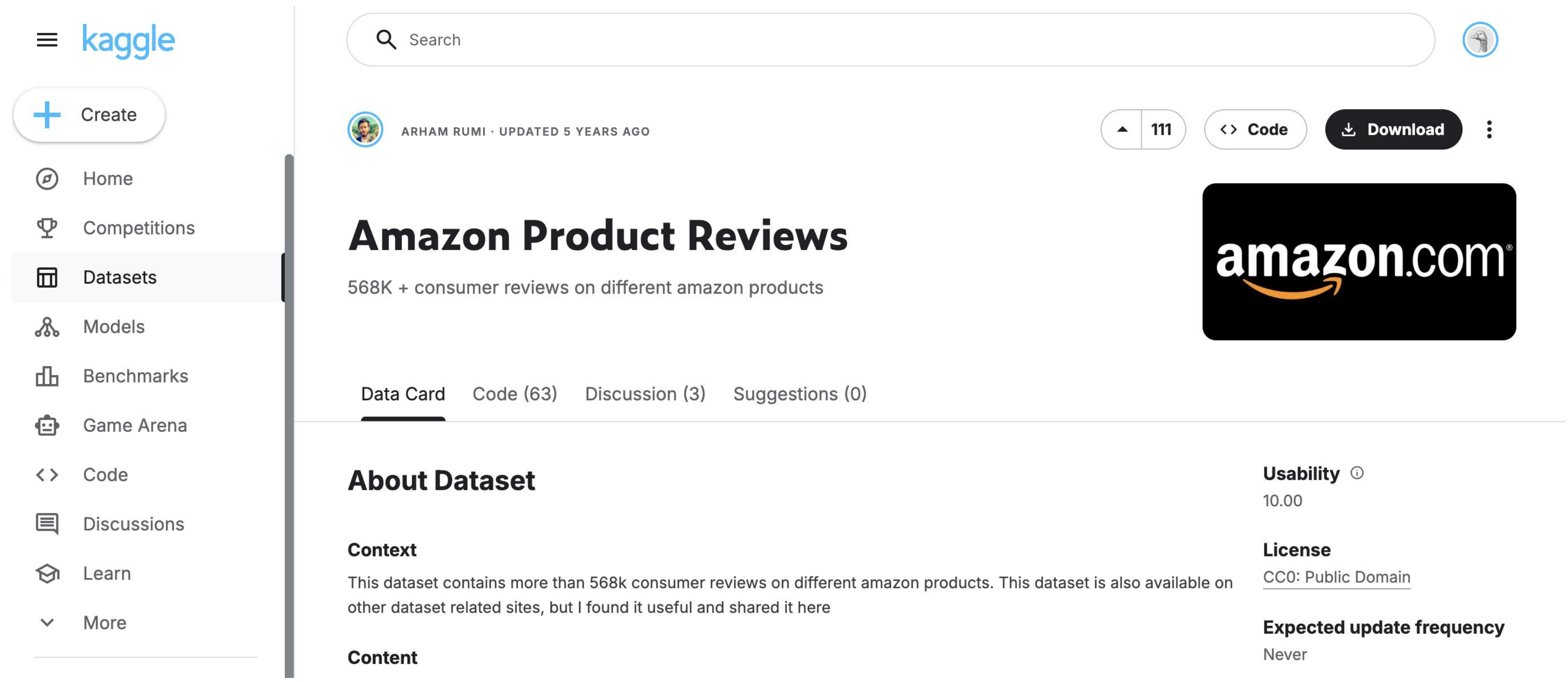
Data Sources and Research Questions

Data Analysis

Findings

Implications and Conclusions

1.2 Data Sources



Data Source: Amazon Product Reviews
Platform: Kaggle
Link: https://www.kaggle.com/datasets/arhamrumi/amazon-product-reviews/data

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2.1 Variable Constructions

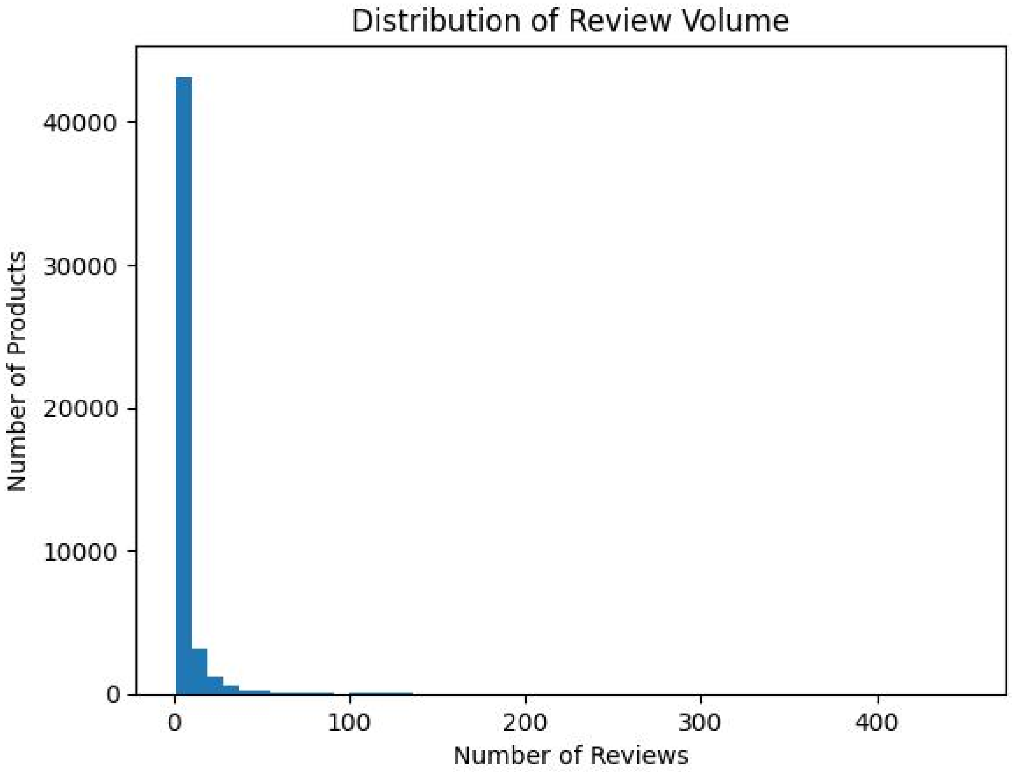
Initial Variables

- ✓ Id: Record ID
- ✓ ProductId: Product ID
- ✓ UserId: User ID who is reviewing
- ✓ ProfileName: User Profile Name who is reviewing
- ✓ HelpfulnessNumerator: Numerator for Helpfulness of review
- ✓ HelpfulnessDenomitor: Denominator for Helpfulness of review
- ✓ Score: Product Rating
- ✓ Time: Review Time in TimeStamp

Constructed Key Variables

- ✓ AvgRating: Average star rating of a product, capturing its overall perceived quality.
- ✓ ReviewCount->LogReviewCount: Natural logarithm of total review count, used as a proxy for product popularity and visibility while reducing skewness.
- ✓ AvgHelpfulness: Average helpfulness ratio of reviews, measuring the perceived informativeness of user-generated content.
- ✓ ProductAge: Time since the first review, capturing cumulative exposure and opportunity to receive reviews.
- ✓ HighReview： Indicator variable equal to 1 for high-review (mature-stage) products and 0 otherwise.

2.2 Descriptive Evidence



Review Volume Is Highly Skewed

Review counts exhibit a highly right-skewed distribution, with most products receiving few reviews and a small fraction accumulating a large number of reviews, motivating a stage-based classification of products.

Given the highly skewed distribution of review volume shown in Figure, we further examine whether the effects of ratings and helpfulness differ across product stages.

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2.3 Regression model

OLS regression with log specification

The model captures how algorithmic reputation signals interact with product maturity to shape review accumulation in an online marketplace.

$$\log(\text{ReviewCount}) = \beta_0 + \beta_1\,\text{AvgRating} + \beta_2\,\text{HighReview} + \beta_3\,(\text{AvgRating} \times \text{HighReview})$$
$$\beta_4\,\text{AvgHelpfulness} + \beta_5\,\text{ProductAge} + \varepsilon$$

Model Split

- When HighReview = 0, the coefficient on AvgRating captures the effect of ratings for low–review (early–stage) products;

$$\log(\text{ReviewCount}) = \beta_0 + \beta_1\,\text{AvgRating} + \beta_4\,\text{AvgHelpfulness} + \beta_5\,\text{ProductAge} + \varepsilon$$

- when HighReview = 1, the effect of ratings equals the sum of the main effect and the interaction term.

$$\log(\text{ReviewCount}) = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\,\text{AvgRating} + \beta_4\,\text{AvgHelpfulness} + \beta_5\,\text{ProductAge} + \varepsilon$$

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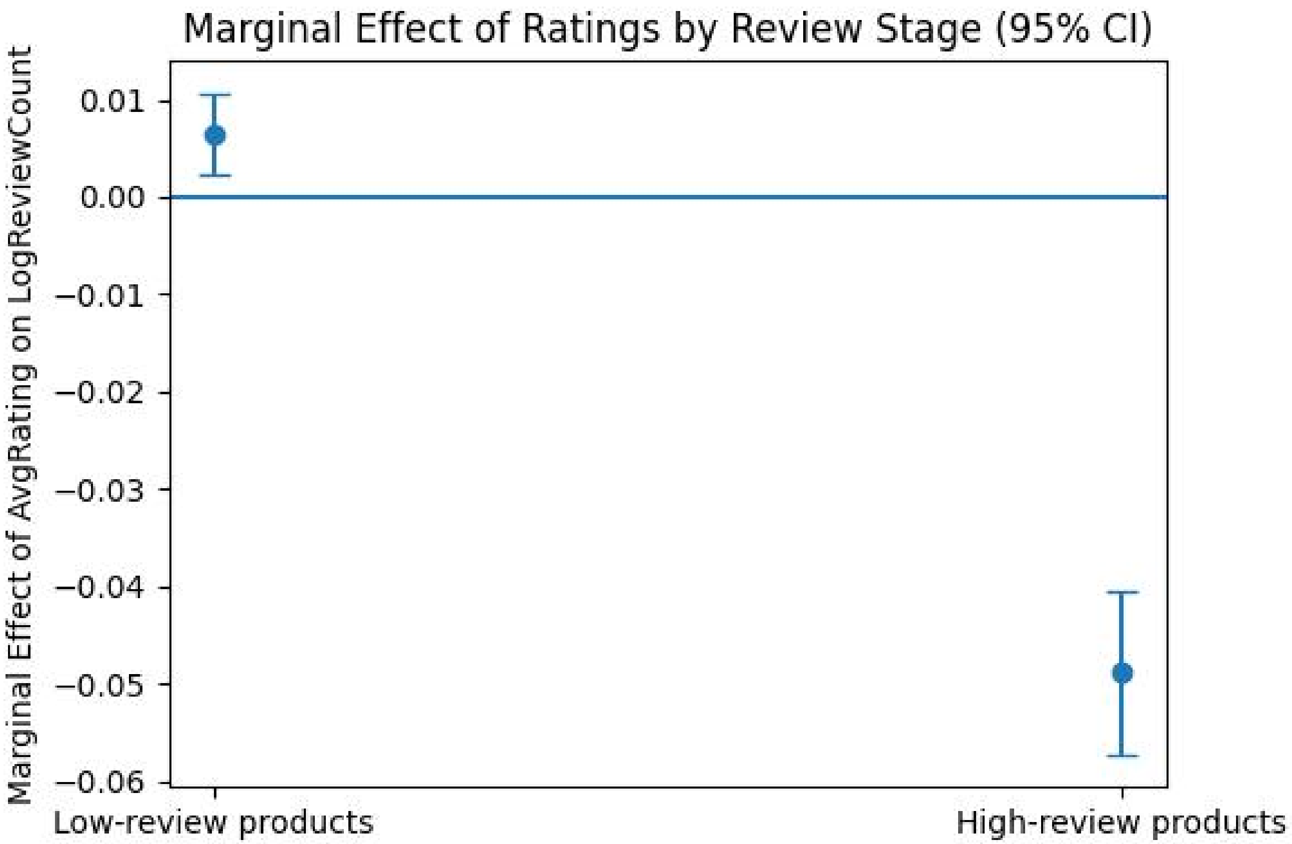
2.4 Marginal Effects Visualization



For low-review products, higher ratings are associated with a small but positive increase in review accumulation. In contrast, for high-review products, the marginal effect of ratings becomes negative and statistically significant, suggesting diminishing or even adverse returns to additional rating improvements once a product reaches a mature review stage.

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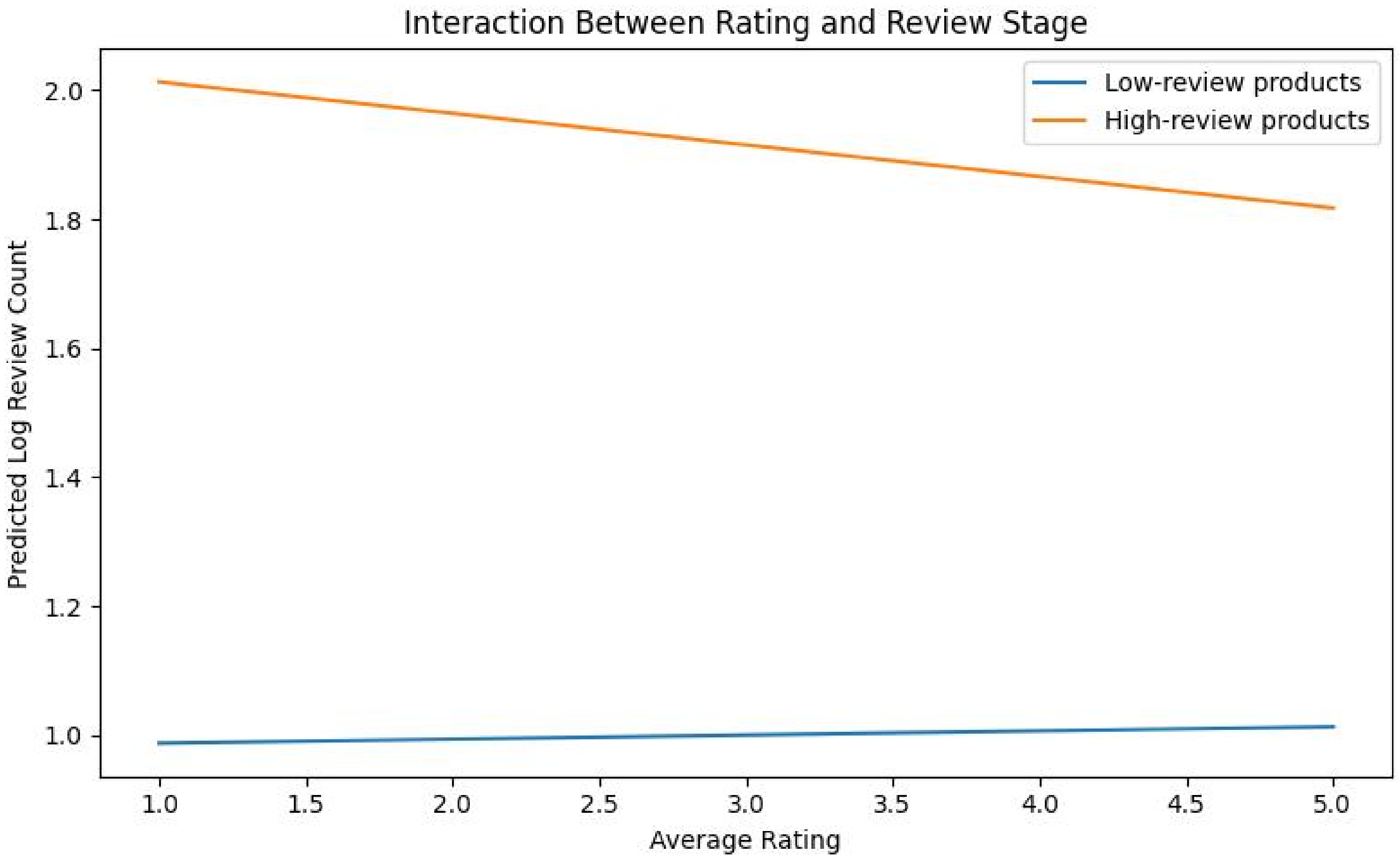
2.5 Interaction Between Rating and Review Stage



The positive effect of ratings on review accumulation is strong for low-review products but flattens substantially for high-review products.

3.1 Key Findings: Stage-Dependent Effects of Ratings

01 Early-Stage Reputation Effect

For low-review products, higher ratings significantly promote additional review accumulation, indicating that ratings function as a strong and informative reputation signal at early stages.

02 Diminishing Marginal Impact at Maturity

As products accumulate a large number of reviews, the marginal effect of ratings weakens and may even turn negative, suggesting diminishing returns to incremental rating improvements.

03 Moderating Role of Review Stage

The effectiveness of ratings is not uniform but depends on the product's review stage, highlighting review volume as a key moderator in algorithmic reputation mechanisms.

4.1 Implications: Platform & Algorithms

Disproportionate Influence of Early Reputation Signals

Ratings play a dominant role in driving visibility and engagement at early stages, implying that small initial differences can lead to large downstream effects through algorithmic amplification.

Risk of Amplifying Early Shocks Under Uniform Rules

Applying uniform ranking and recommendation rules across all products may unintentionally magnify early positive or negative shocks,

Benefits of Stage-Aware Rating Weighting

Incorporating review stage into algorithmic weighting—such as down-weighting ratings for mature products—may improve fairness and information

Our findings suggest that reputation signals interact with product maturity, raising important considerations for how platforms design and weight rating-based algorithms.

4.2 Implications: Sellers / Firms

The stage–dependent effectiveness of ratings also has important strategic implications for how sellers manage reputation and engagement over the product lifecycle.

01

Prioritize Early Reputation Building

For new or low–review products, investing in quality, service, and early customer engagement is critical, as ratings exert a disproportionately large impact on subsequent review growth and visibility.

02

Shift Focus Beyond Ratings at Maturity

Once products accumulate abundant reviews, marginal improvements in ratings yield limited returns, suggesting that sellers should shift attention toward pricing, differentiation, or promotional strategies.

03

Adopt Stage–Specific Reputation Strategies

A one–size–fits–all approach to reputation management is suboptimal; firms benefit from dynamically adjusting their strategies as products transition from early to mature stages.

4.3 Conclusions

01 Stage–Dependent Role of Ratings

Ratings significantly influence review accumulation, but their impact is highly stage–dependent

02 Early–Stage Dominance of Reputation Signals

Reputation signals are most effective for low–review products, while marginal effects diminish as products reach maturity.

03 Algorithmic and Market Implications

Interaction–based analysis reveals how platform algorithms and market dynamics shape reputation outcomes

04 Conlusions

Overall, the findings emphasize the importance of incorporating product lifecycle considerations into the design and evaluation of online reputation systems.

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I sincerely welcome your criticism and corrections.

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Appendix

```
Run Cell | Run Below | Debug Cell
# %%
import kagglehub
import shutil
import os
import pandas as pd
import numpy as np
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt
src_path = kagglehub.dataset_download("arhamrumi/amazon-product-reviews")

Run Cell | Run Above | Debug Cell
# %%
dst_path = os.getcwd()
Run Cell | Run Above | Debug Cell
# %%
for file in os.listdir(src_path):
    shutil.copy(os.path.join(src_path, file), dst_path)

Run Cell | Run Above | Debug Cell
# %%
print("Dataset copied to:", dst_path)
Run Cell | Run Above | Debug Cell
# %%
df = pd.read_csv("Reviews.csv")
Run Cell | Run Above | Debug Cell
# %%
print(df.head())
Run Cell | Run Above | Debug Cell
```

```
df = df[[
    "ProductId",
    "Score",
    "HelpfulnessNumerator",
    "HelpfulnessDenominator",
    "Time"
]]
df = df.dropna(subset=["ProductId", "Score"])
df = df[df["HelpfulnessDenominator"] != 0]
df["HelpfulnessRatio"] = np.where(
    df["HelpfulnessDenominator"] > 0,
    df["HelpfulnessNumerator"] / df["HelpfulnessDenominator"],
    np.nan
)
Run Cell | Run Above | Debug Cell
# %%
product_df = df.groupby("ProductId").agg(
    AvgRating=("Score", "mean"),
    ReviewCount=("Score", "count"),
    AvgHelpfulness=("HelpfulnessRatio", "mean"),
    FirstReviewTime=("Time", "min"),
    LastReviewTime=("Time", "max")
).reset_index()
print(product_df)
Run Cell | Run Above | Debug Cell
# %%
# Sales proxy (log 形式)
product_df["LogReviewCount"] = np.log1p(product_df["ReviewCount"])
# Product age (以“年”为单位)
product_df["ProductAge"] = (
    product_df["LastReviewTime"] - product_df["FirstReviewTime"]
) / (60 * 60 * 24 * 365)
product_df
```

Appendix

```python
# %%
plt.figure()
plt.hist(product_df["ReviewCount"], bins=50)
plt.xlabel("Number of Reviews")
plt.ylabel("Number of Products")
plt.title("Distribution of Review Volume")
plt.show()
Run Cell | Run Above | Debug Cell
# %%
median_reviews = product_df["ReviewCount"].median()
product_df["HighReview"] = (product_df["ReviewCount"] > median_reviews).astype(int)
print(product_df["HighReview"])
Run Cell | Run Above | Debug Cell
# %%
model_2 = smf.ols(
    formula="""
    LogReviewCount ~ AvgRating * HighReview + AvgHelpfulness + ProductAge
    """,
    data=product_df
).fit()
print(model_2.summary())
# extract coefficients
beta_rating = model_2.params["AvgRating"]
beta_interaction = model_2.params["AvgRating:HighReview"]
# marginal effects
marginal_low = beta_rating
marginal_high = beta_rating + beta_interaction
# dataframe for plotting
me_df = pd.DataFrame({
    "Review Stage": ["Low-review products", "High-review products"],
    "Marginal Effect": [marginal_low, marginal_high]
})
```

```python
# dataframe for plotting
me_df = pd.DataFrame({
    "Review Stage": ["Low-review products", "High-review products"],
    "Marginal Effect": [marginal_low, marginal_high]
})
print(me_df)

Run Cell | Run Above | Debug Cell
# %%
# coefficients
beta1 = model_2.params["AvgRating"]
beta3 = model_2.params["AvgRating:HighReview"]
# covariance matrix
vcov = model_2.cov_params()
var_beta1 = vcov.loc["AvgRating", "AvgRating"]
var_beta3 = vcov.loc["AvgRating:HighReview", "AvgRating:HighReview"]
cov_13 = vcov.loc["AvgRating", "AvgRating:HighReview"]
# marginal effects
me_low = beta1
me_high = beta1 + beta3
# standard errors
se_low = np.sqrt(var_beta1)
se_high = np.sqrt(var_beta1 + var_beta3 + 2 * cov_13)
ci_low = (me_low - 1.96 * se_low, me_low + 1.96 * se_low)
ci_high = (me_high - 1.96 * se_high, me_high + 1.96 * se_high)

me_df = pd.DataFrame({
    "Review Stage": ["Low-review products", "High-review products"],
    "Marginal Effect": [me_low, me_high],
    "CI Lower": [ci_low[0], ci_high[0]],
    "CI Upper": [ci_low[1], ci_high[1]]
})
```